

Multilingual Customer Chat Translator & Responder — Requirements Sheet

Owner: [Name] | Date: October 05, 2026 | Version: 1.0 | Status: Draft

1. Overview

The Multilingual Customer Chat Translator & Responder enables support agents to handle customer conversations in any language without requiring multilingual staff. It automatically detects language, translates messages, analyzes intent and sentiment, and drafts brand-aligned replies for agent review. This reduces response time, improves customer satisfaction, and scales support operations efficiently.

2. Preface

Problem: Support teams struggle with multilingual customers, leading to delayed responses, inconsistent quality, and higher costs from specialized agents.
Solution: An AI-powered system that detects language, translates in real time, analyzes context, and generates policy-compliant replies with agent oversight.
Goal: Enable English-speaking agents to deliver accurate, on-brand support in any language while maintaining control, auditability, and compliance.

3. Requirements

3.1 Functional

ID	Requirement
FR1	Auto-detect customer language.
FR2	Translate inbound message into agent language.
FR3	Display original + translation side-by-side.
FR4	Detect intent, sentiment, urgency.
FR5	Generate reply in customer's language using brand tone and policy.
FR6	Provide back-translation for agent verification.
FR7	Allow agent to approve, edit, or reject reply.
FR8	Escalate refunds, legal, safety, angry, low-confidence cases.
FR9	Maintain glossary for product names, legal terms, brand voice.
FR10	Log original, translation, reply, language, confidence, agent action.
FR11	Support auto-responder mode with confidence threshold.

3.2 Non-Functional

- **Latency:** Translation < 2s; suggested reply < 5s.
- **Accuracy:** > 90% meaning preservation; > 95% language detection.
- **Availability:** 99.9% uptime.
- **Security:** TLS, encryption at rest, RBAC, PII redaction.
- **Compliance:** GDPR / CCPA, configurable retention.
- **Scale:** [X] concurrent chats, [Y] languages.

3.3 Integrations: Chat platform, CRM, ticketing, translation API/LLM, SSO, analytics.

3.4 Workflow: Customer message → detect language → translate → analyze → draft reply → agent review → send → log.

3.5 Acceptance Criteria

- Agent reads any supported language in English.
- Customer receives reply in their own language.
- Back-translation available before sending.
- Escalation triggers correctly.
- Audit log complete.
- Latency and accuracy thresholds met.

3.6 Success Metrics: Translation accuracy, first response time, resolution time, CSAT, escalation rate, auto-send accuracy.

3.7 Assumptions & Constraints

- LLM/translation API available.
- Agent language is English.
- Human approval required for MVP.
- No legal certification.

3.8 Approvals: Product: _____ Engineering: _____ Support: _____ Compliance: _____